

Loops

Used to execute a block of code repeatedly until a condition is met.

↙
for loop

↘ while loop.

for i in range (0, 10) :
do this → print(i)

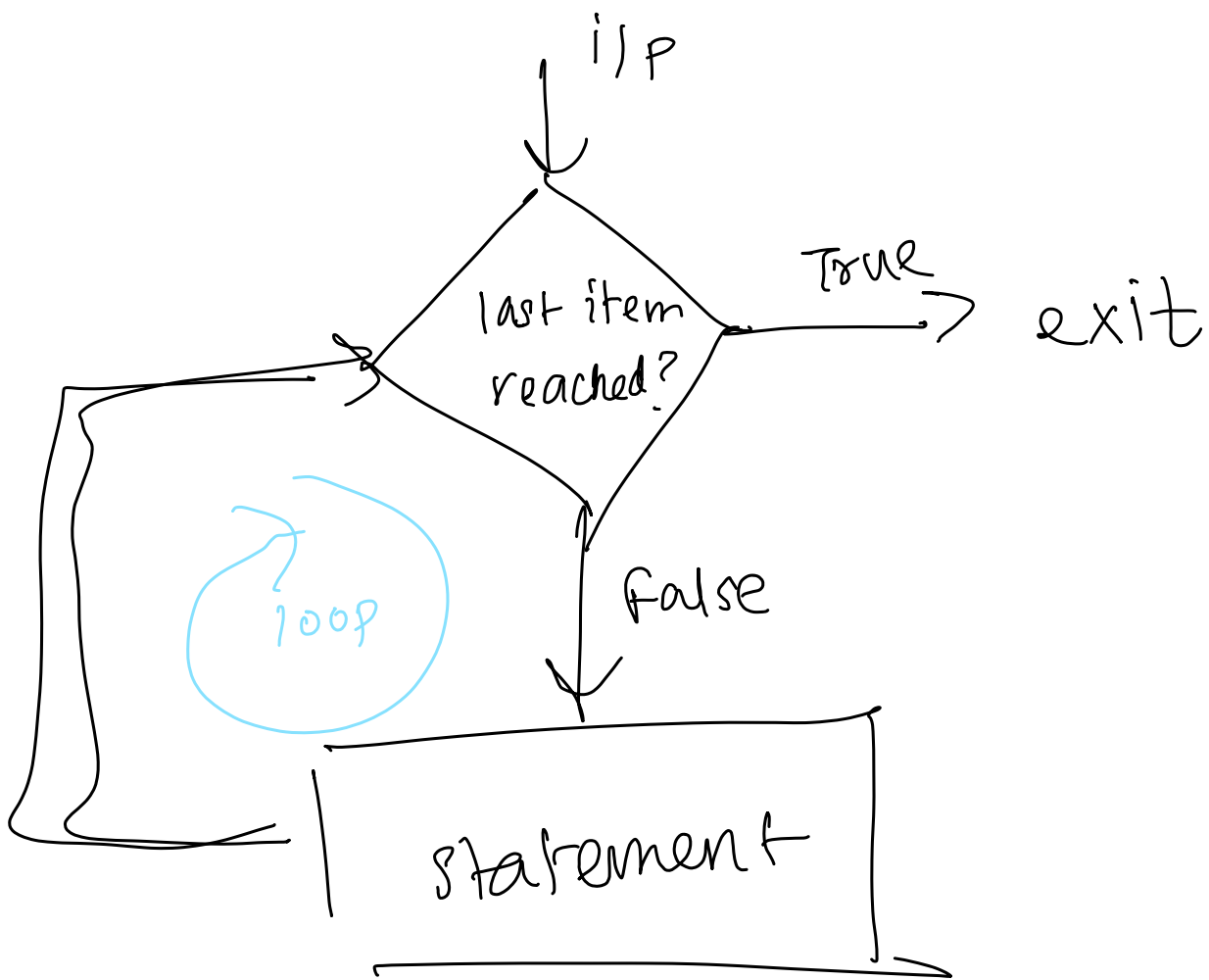
$$0 \leq i < 10$$

0 1 2 3 4 5 6

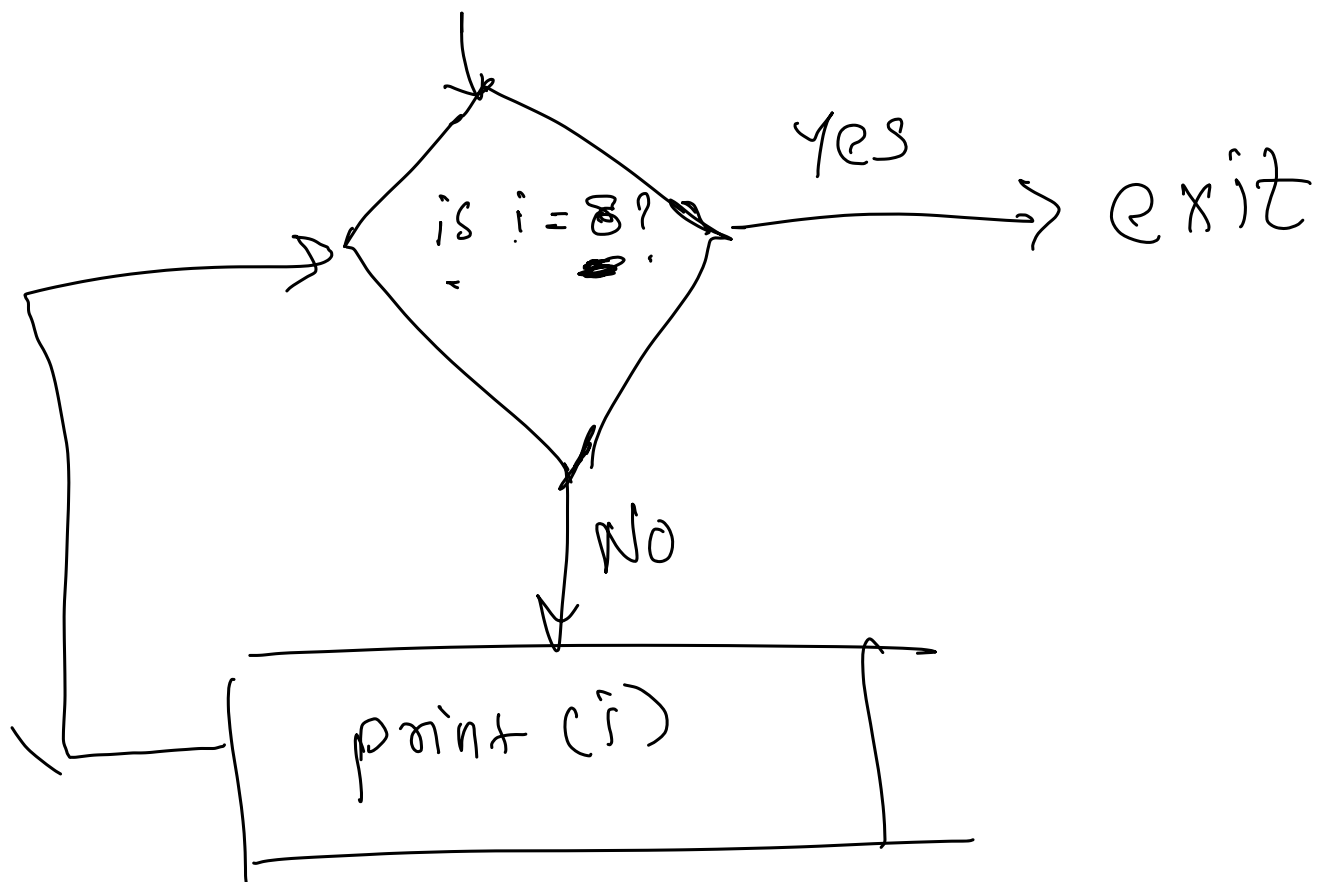
7 8 9

i in range (0, 8)

0 ~~≤~~ i < 8 start ≤ i < end



range(0, 1, 2, 3, 4, 5, 6, 7, 8)
i = 1



i	O/P
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	exit

$a = [\overset{0}{\text{"apples"}}, \overset{1}{\text{"oranges"}}, \overset{2}{\text{"mangoes"}}]$

`print (a[1])` $\Rightarrow$ "oranges"

length of the list / size of the list
= number of items in the list

$$\text{den}(a) = 3$$

len()
predefined function.

a = [⁰"apples", ¹"oranges", ²"mangoes"]

```
print(a[0])
```

```
print (a[1])
```

```
print (a (2))
```

$$(0, 1, 2)$$

start $\leq i \leq$ end

for i in range (0, 3):

print (a[i])

0, 1, 2

for i in range (0, len(a)):

print (a[i])

While loop

while (condition is True) :

do this

count = 0

while (count < 3):

~~count = count + 1~~

~~print (count)~~

count	count	Output
0	1	1
1	2	2
2	3	3
3	exit	exit

1, 2, 3

count = 0

while count < 3 :

print(count)

count = count + 1

Q. 12

count	print	count
0	0	1
1	1	2
2	2	3
3	exit	exit

Qmp

Dangerous

count = 0

True

while count < 3:
 print(count)

infinite
loop

count

print

0

0

0

0

0

0

0

0

1

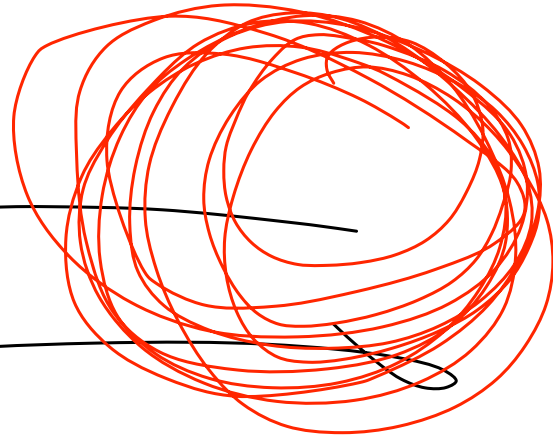
1

1

1

1

1



computation

crash

* Always have
an exit condition.

Infinite while loop:

Runs continuously because the condition always remains True.

```
while 1 == 1:
```

```
    print ("Hahaha")
```

```
while True:
```

```
    print (True)
```

class example

count table

count	print	count
1	1	2
2	2	3
3	3	4
4	4	5
5	5	6
6	exit	exit

Total # iterations of this loop = 6

while loop backward example

num	print	num ⁺
5	5	4
4	4	3
3	3	2
2	2	1
1	1	0
0	exit	exit
"Blast off"		

num = 1

total = 0 ← sum

while total ≤ 10 :

total = total + num

num = num + 1

num	total	total + num	num + 1
1	0	1	2
2	1	1 + 2 = 3	3
3	3		

```
for i in range (1,6):  
    print ("*" * i)
```

i	print output
1	*
2	* *
3	* * *
4	* * * *
5	* * * * *
6	exit